

KYC & AML Compliance Project

Customer Risk Assessment & Transaction Monitoring Framework

Domain: Banking & Financial Services Compliance | Tool: Microsoft Excel | Duration: 4 Weeks | Role: KYC/AML Analyst (Fresher)

75	150	26	8	5	6
Customer Profiles	Transactions Monitored	Suspicious Alerts	SARs / STRs Filed	Compliance Sheets	Regulatory Frameworks

PROJECT OVERVIEW

Developed a comprehensive KYC (Know Your Customer) and AML (Anti-Money Laundering) compliance framework in Microsoft Excel, simulating real-world banking operations for a mid-sized Indian private bank. The project covers the complete compliance lifecycle — from customer onboarding and risk scoring to transaction monitoring, alert classification, and regulatory reporting — fully aligned with RBI KYC Master Directions 2016, PMLA 2002, FATF 40 Recommendations, and FIU-IND reporting guidelines.

OBJECTIVES

- Perform Customer Due Diligence (CDD) for 75 diverse customer profiles across multiple risk segments
- Build a weighted risk scoring model (0–100) using PEP status, occupation, income & source of funds
- Monitor 150 transactions across 8 transaction types for suspicious patterns
- Classify alerts as Normal / Low / Medium / High using rule-based AML logic
- Generate SAR/STR filings for high-risk transactions and report to FIU-IND within 3-day SLA
- Create a live Compliance Dashboard with 6 KPIs tracked through cross-sheet Excel formulas
- Maintain full audit trail of CDD decisions, alert reasoning, and regulatory actions

REGULATORY FRAMEWORK

PMLA 2002 — Prevention of Money Laundering Act obligations & reporting
RBI KYC Master Directions 2016 — Customer identification & ongoing monitoring
FATF R.12 — Enhanced measures for Politically Exposed Persons (PEPs)
FATF R.26 — Supervision of high-risk business categories
FIU-IND Guidelines — STR/CTR filing thresholds and timelines
Basel AML Index — Country risk scoring for cross-border transactions
SEBI KYC Norms — KRA registration and periodic update requirements

RISK SCORING METHODOLOGY

Parameter	Criteria	Score	Max	Reference
PEP Status	Politically Exposed Person	40	40	FATF R.12 / PMLA S.12
PEP Status	Close Associate of PEP	30	40	FATF R.12
PEP Status	Non-PEP	0	40	FATF R.12
Occupation	High-Risk (Import-Export, Jewellery, Forex, Crypto)	25	25	FATF R.26
Occupation	Medium-Risk (Business Owner, Contractor, Broker, Dealer)	15	25	RBI KYC Dir.
Occupation	Low-Risk (Salaried, Professional, Student, Pensioner)	0	25	RBI KYC Dir.
Annual Income	Rs. 50 Lakh and above	20	20	FIU-IND Circular
Annual Income	Rs. 20 Lakh – Rs. 50 Lakh	10	20	FIU-IND Circular

Parameter	Criteria	Score	Max	Reference
Annual Income	Rs. 10 Lakh – Rs. 20 Lakh	5	20	FIU-IND Circular
Annual Income	Below Rs. 10 Lakh	0	20	FIU-IND Circular
Source of Funds	High-Risk (Trade Revenue, Donations, Exchange Margin)	10	10	PMLA 2002 S.3
Source of Funds	Standard (Salary, Professional Fees, Pension)	0	10	PMLA 2002 S.3
ID Verification	Incomplete / Documents Pending	5	5	RBI KYC Dir. S.16
ID Verification	Full KYC Completed	0	5	RBI KYC Dir. S.16
Country Risk	FATF Grey-List / High-Risk Jurisdiction	5	5	FATF Public Stmt.
Country Risk	Standard / Low-Risk Country	0	5	FATF Public Stmt.
TOTAL SCORE			100	

LOW (0 – 19)	MEDIUM (20 – 44)	HIGH (45 – 69)	VERY HIGH (70 – 100)
Simplified DD Quarterly Review Annual Update	Standard DD Monthly Review 6-Month Update	Enhanced DD Weekly Review 3-Month Update	EDD + Ongoing SAR Daily Monitoring 1-Month Update

CUSTOMER RISK PROFILE — SAMPLE DATA (15 of 75)

Cust ID	Name	Occupation	Income (Rs.)	PEP	Risk Score	Risk Category	CDD Level	Status
C001	Rahul Sharma	Salaried Employee	6,00,000	No	5	Low	Simplified DD	Approved
C003	Amit Verma	Politician	36,00,000	Yes	75	Very High	EDD + SAR	Under Review
C005	Raj Mehta	Import-Export Trader	60,00,000	No	55	High	Enhanced DD	Under Review
C007	Karan Shah	Management Consultant	14,40,000	No	20	Medium	Standard DD	Approved
C011	Deepak Sharma	Real Estate Agent	24,00,000	No	25	Medium	Standard DD	Approved
C017	Manish Tiwari	Politician	48,00,000	Yes	80	Very High	EDD + SAR	Under Review
C019	Rohit Desai	Forex Trader	36,00,000	No	50	High	Enhanced DD	Under Review
C027	Sanjay Banerjee	Jewellery Trader	54,00,000	No	45	High	Enhanced DD	Pending Docs
C031	Nikhil Aggarwal	Import-Export Trader	72,00,000	No	60	High	Enhanced DD	Under Review
C033	Pankaj Dubey	Money Changer	36,00,000	No	50	High	Enhanced DD	Pending Docs
C039	Dinesh Bhat	Politician	60,00,000	Yes	85	Very High	EDD + SAR	Under Review
C045	Mukesh Gupta	Textile Trader	48,00,000	No	45	High	Enhanced DD	Under Review
C049	Bharat Patel	Import-Export Trader	84,00,000	No	65	High	Enhanced DD	Under Review
C052	Priya Nair	Software Engineer	18,00,000	No	5	Low	Simplified DD	Approved

Cust ID	Name	Occupation	Income (Rs.)	PEP	Risk Score	Risk Category	CDD Level	Status
C068	Sunita Bhatt	NGO / Trust Worker	7,20,000	No	15	Low	Simplified DD	Approved

AML TRANSACTION MONITORING FRAMEWORK

Alert Trigger Rules:

- CTR — Cash deposit / withdrawal exceeding Rs. 5 Lakh in single transaction
- STR — Any transaction by PEP exceeding Rs. 2 Lakh regardless of channel
- Large Value Alert — RTGS / Wire Transfer exceeding Rs. 10 Lakh
- Cross-Border Alert — Transfer to FATF grey-list country exceeding Rs. 3 Lakh
- Cash Threshold — Cash transactions in Rs. 2–5 Lakh range for monitoring
- Velocity Alert — More than 5 transactions by same customer in 7 days
- Income Mismatch — Transaction amount exceeds 20% of monthly income profile

Monitoring Methodology Applied:

- Customer income profiling — every transaction benchmarked vs. declared income
- Geography risk scoring — UAE, Singapore, Mauritius, BVI flagged as high-risk
- Transaction velocity check — frequency and clustering pattern analysis
- Cash vs. digital ratio — disproportionate cash use flagged for review
- Round-number analysis — transactions ending in 000 or 00000 investigated
- SAR auto-trigger — all High-level alerts generate mandatory SAR filing
- 3-day SLA tracking — SAR filed within 72 hours of alert confirmation

SAMPLE HIGH-RISK TRANSACTIONS IDENTIFIED (12 of 26 Alerts)

Txn ID	Customer	Txn Type	Amount (Rs.)	Alert Type	Level	Action Taken
T023	Amit Verma (PEP, Politician)	RTGS Transfer	4,50,000	STR	HIGH	SAR filed to FIU-IND within 24 hrs
T047	Raj Mehta (Import-Export)	Cash Deposit	12,00,000	CTR	HIGH	CTR + SAR filed; Source of funds sought
T061	Manish Tiwari (PEP)	Wire Transfer	3,00,000	STR	HIGH	SAR filed; Account placed under watch
T078	Sanjay Banerjee (Jewellery)	Cash Deposit	8,50,000	CTR	HIGH	CTR filed; EDD initiated
T085	Nikhil Aggarwal (Import)	RTGS to UAE	9,00,000	Cross-Border	MEDIUM	Cross-border review; KYC refreshed
T092	Bharat Patel (Import)	Wire Transfer	15,00,000	Large Value + STR	HIGH	SAR filed; Transaction frozen pending review
T103	Dinesh Bhat (PEP, Politician)	NEFT Transfer	2,50,000	STR	HIGH	SAR filed to FIU-IND
T117	Pankaj Dubey (Money Changer)	Cash Withdrawal	6,00,000	CTR	HIGH	CTR filed; Business verification initiated
T124	Rohit Desai (Forex Trader)	Wire to Mauritius	5,50,000	Cross-Border	MEDIUM	Cross-border review; Geography flagged
T136	Mukesh Gupta (Textile Trader)	Cash Deposit	7,20,000	CTR	HIGH	CTR + SAR filed; Velocity check triggered
T141	Unknown Referral (New Account)	IMPS Deposit	3,80,000	Income Mismatch	MEDIUM	Income profile review; Source queried
T149	Kiran Seth (Business Owner)	5x Deposits/week	9,10,000	Velocity Alert	MEDIUM	Velocity pattern reviewed; STR under eval.

EXCEL SKILLS & FUNCTIONS DEMONSTRATED

Excel Function / Feature	How It Was Applied in This Project
COUNTIF / COUNTIFS	Count customers by risk category, alert level, CDD type, and status across all 5 sheets
IF / Nested IF	Risk classification — categorise customers into Low/Medium/High/Very High based on score
VLOOKUP / XLOOKUP	Cross-reference Customer IDs between KYC sheet and AML transaction sheet
SUM / SUMIF / SUMIFS	Aggregate transaction amounts by customer, alert type, transaction type, and time period
Cross-Sheet Formulas	Live links from KYC_Customer_Data and AML sheets to Compliance Dashboard KPIs
Conditional Formatting	Colour-code risk levels — Green (Low), Yellow (Medium), Red (High), Dark Red (Very High)
Data Validation	Dropdown lists for Risk Category, CDD Level, Alert Type, SAR Status, and ID Doc Type
Pivot Tables	Summarise 150 transactions by customer, month, alert type, channel, and transaction type
IFERROR	Wrap all VLOOKUP/division formulas to prevent #N/A and #DIV/0! errors in dashboard
TEXT / DATE functions	Format transaction dates for monthly trend analysis and SLA breach calculation

KEY OUTCOMES & FINDINGS

- ✓ Identified **9 High-risk and Very High-risk customers** (PEPs and high-volume traders) requiring Enhanced Due Diligence
- ✓ Flagged **26 suspicious transactions** out of 150 monitored — an alert rate of **17.3%**, aligned with industry norms
- ✓ Filed **8 SARs/STRs to FIU-IND** within the mandatory 72-hour reporting window, with full audit trail maintained
- ✓ Detected **cross-border exposure** in 11 transactions involving UAE, Singapore, Mauritius, and BVI jurisdictions
- ✓ Achieved **84% customer approval rate**, 10% under enhanced review, 6% pending document verification
- ✓ Built a **live Compliance Dashboard** tracking 6 KPIs in real time using 55+ cross-sheet Excel formulas
- ✓ Modelled **income mismatch detection** — flagging 4 cases where transaction value exceeded monthly income profile by 3x

PROJECT FILE STRUCTURE — 5 EXCEL SHEETS

Sheet	Name	Contents
Sheet 1	KYC_Customer_Data	75 customer profiles with Name, DOB, Occupation, Income, PEP Status, ID Verification, Risk Score (formula-driven), Risk Category, CDD Level, and Approval Status
Sheet 2	AML_Transaction_Monitor	150 transactions covering Cash Deposit, RTGS, NEFT, Wire Transfer, IMPS, Cheque Deposit, ATM Withdrawal with auto-calculated Alert Type and Alert Level
Sheet 3	Risk_Scoring_Matrix	Full parameter weight table with score ranges, CDD requirements, review frequencies, and regulatory reference for each parameter — used as methodology documentation
Sheet 4	SAR_STR_Report_Log	All 8 SARs/STRs with reference numbers, filing dates, transaction details, reason for filing, reporting authority (FIU-IND), and current status
Sheet 5	Compliance_Dashboard	Live KPI dashboard with 6 metrics auto-populated via COUNTIF and cross-sheet formulas: Approval Rate, Alert Rate, SAR Rate, Risk Distribution, PEP Exposure, CTR Rate